



2026 oneM2M International Hackathon

Overview of the Hackathon

October 5th, 2026

SeungMyeong Jeong / SDS WG Vice-chair

Bob Flynn / TDE WG Chair

oneM2M, the IoT Global Standard



Over 250 member organizations
in oneM2M



Regional Standards Development organizations SDO's, are converting oneM2M technical specifications into normative Standards. Those standards are then references by regional legal and regulatory bodies



Release 2 transposition
ITU-T SG20 Y.4500.x

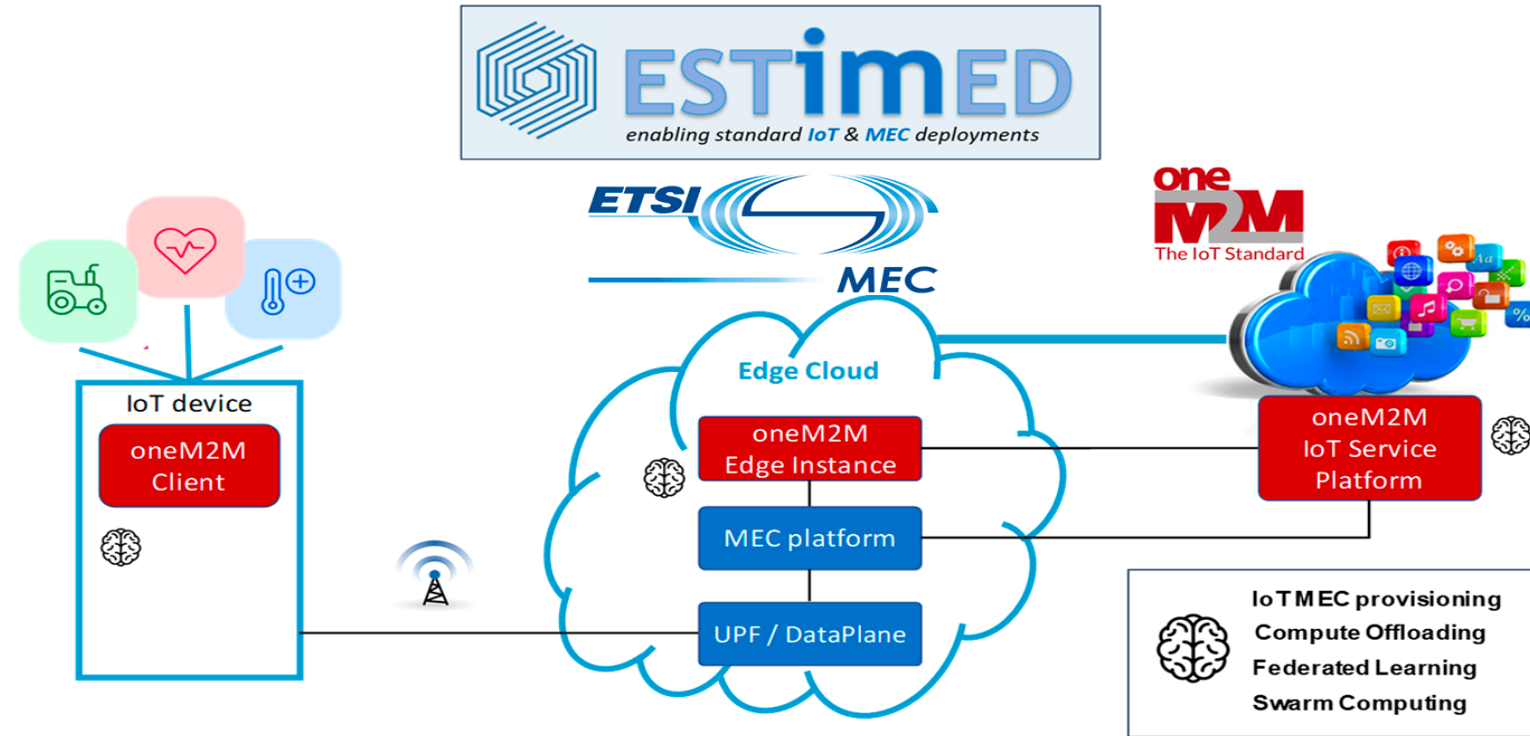
oneM2M as ITU-T Recommendations



- Technical specifications and reports have been transposed
- oneM2M will have better visibility and wider adoption globally

Recommendation	Title	oneM2M TS
Y.4500.1	Functional Architecture	TS-0001
Y.4500.2	Requirements	TS-0002
Y.4500.3	Security Solutions	TS-0003
Y.4500.4	Core Protocol	TS-0004
Y.4500.5	Management Enablement (OMA)	TS-0005
Y.4500.6	Management Enablement (BBF)	TS-0006
Y.4500.8	CoAP Protocol Binding	TS-0008
Y.4500.9	HTTP Protocol Binding	TS-0009
Y.4500.10	MQTT Protocol Binding	TS-0010
Y.4500.11	Common Terminology	TS-0011
Y.4500.12	Base Ontology	TS-0012
Y.4500.13	Interoperability Testing	TS-0013
Y.4500.14	LwM2M Interworking	TS-0014
Y.4500.15	Testing Framework	TS-0015
Y.4500.20	WebSocket Protocol Binding	TS-0020
Y.4500.22	Field Device Configuration	TS-0022
Y.4500.23	Home Appliances Information Model and Mapping	TS-0023
Y.4500.32	MAF and MEF Interface Specification	TS-0032

MEC and oneM2M



- ESTIMED: Enabling Standardized IoT deployments in MEC Environments for advanced systems
- MEC: Multi-access Edge Computing



Co-funded by
the European Union



The ESTIMED project brings together open standards in ETSI ISG MEC and in oneM2M on a horizontal platform for interoperable and scalable IoT systems.

Goal of 2026 oneM2M Hackathon



Design and develop a game-changing **IoT solution** that addresses pressing environmental and societal challenges. Whether it's improving urban living, conserving natural resources, or creating a smarter, more sustainable world, your solution has the power to make a lasting impact.

Harness the capabilities of **oneM2M** and turn your ideas into solutions that can drive real change.

Build your solution using any of the available **oneM2M CSE** implementations and bring your vision for a smarter, more connected world to life. Prototype EDGE-IoT solutions that interwork with the ETSI MEC standard.

Explore **AI + IoT (AIoT)** applications using the AI capabilities being standardized in **oneM2M Release 5** and **ETSI MEC**.

Open-source oneM2M platforms

- Members support different projects for different dev environments
 - Different open sources gives better opportunity for standard adoption

				
Lead		Andreas Kraft	 세종대학교 SEJONG UNIVERSITY	
Homepage	https://github.com/iotketi/mobius4	https://acmecse.net	https://github.com/seslabSJU/tinyIoT	https://www.eclipse.org/om2m
License	BSD 3-Clause	BSD 3-Clause	BSD 3-Clause	EPL 1.0
Offering	Platform, Resource Browser	Platform, Edge, Resource Browser	Platform & Edge	Platform
Binding	HTTP, MQTT	HTTP, MQTT, CoAP, WebSocket	HTTP, MQTT, CoAP, WebSocket	HTTP, CoAP
Format	JSON	JSON, CBOR	JSON	XML, JSON
Language	Javascript (Node.js)	Python	C	Java (OSGi)

ETSI MEC Sandbox (open-source)



Multi-Access Edge Computing

Run your app beside the base station, not in a distant cloud.

ETSI MEC moves compute to the edge of the mobile network (next to 4G/5G towers or base stations). Your app gets real-time location & network data responding in milliseconds, not seconds.

Ultra-low latency
ms, not seconds

Location-aware
Real-time device tracking

Network-aware
Signal, load, handovers

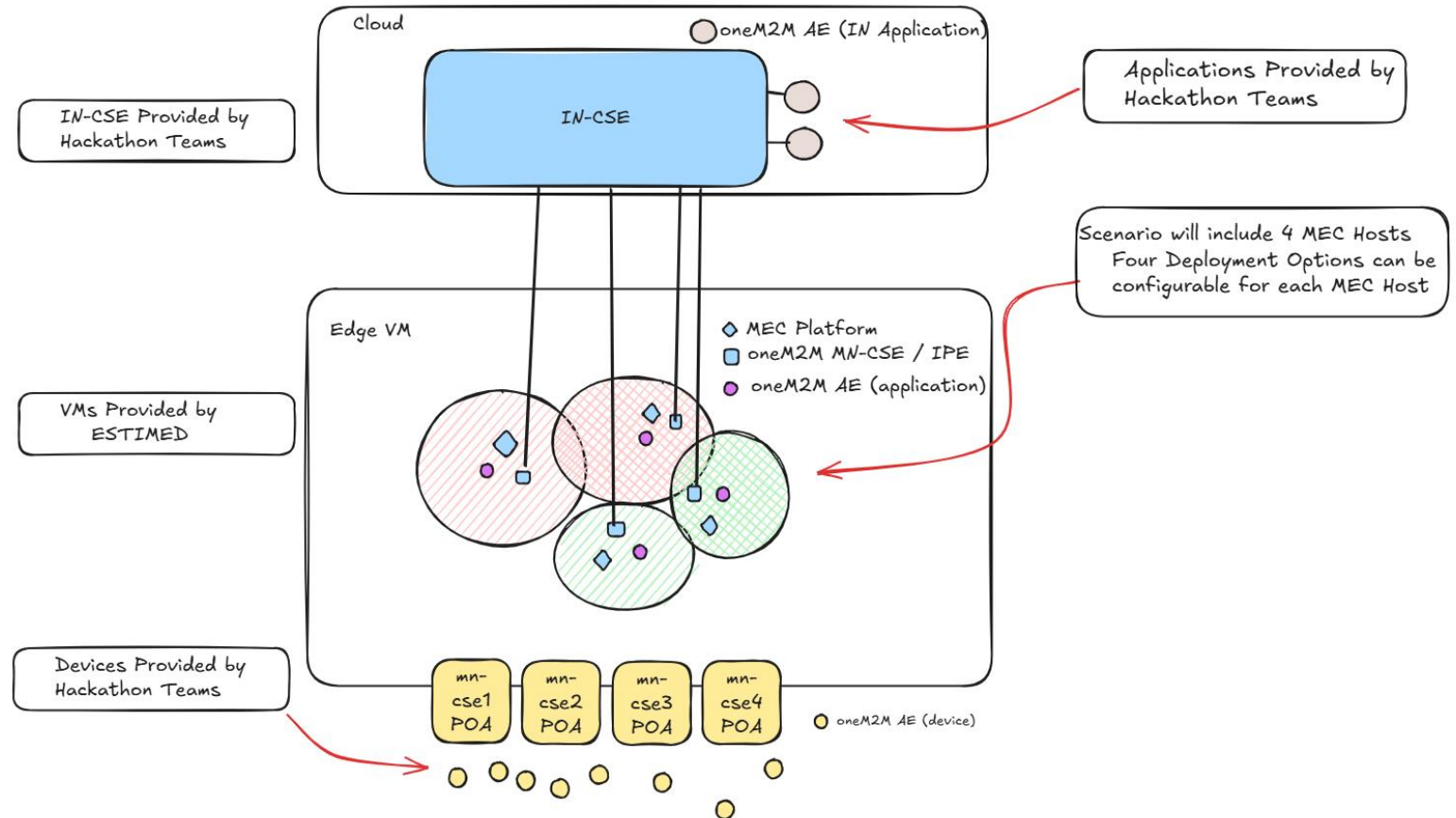
Open REST APIs
Free sandbox to test



Lead By: ETSI ISG MEC

Useful links:

- [ETSI MEC Sandbox user interface guide](#)
- [ETSI MEC Sandbox Deployment](#)



ETSI GR MEC-DEC 050 - ESTIMED Use Cases & Recommendations

oneM2M TR-0077 oneM2M and MEC integration scenario and mechanisms

ETSI GR MEC-DEC 051 - Multi-access Edge Computing (MEC); oneM2M and MEC interworking and deployments

Hackathon schedule



- Deliverables submission by November 20th
 - Document on hackster.io (w/ demo video)
 - Source codes on github
- Judges review the deliverables to pick winners
- Evaluation will be 30 mins session for each team
 - Presentation and QnA



Deliverables of the projects



Each team will produce

- A hackster.io project that is a clear guide on building your IoT solution. It includes
 - Description of the issue and how your project addresses it
 - An explanation of the oneM2M features used in your solution
 - A description of the devices and software used (oneM2M platform, sensors, actuator, simulators, etc ...)
- A public github repository to release the source codes
- A short demo video (< 5 min) showing the solution in action



document sample
of a previous winner



[oneM2M Autonomous cooperative smart delivery system](#)

Jang Hyochang, Jeong Hyeryeong, Kim Sanggun, Soorim Yang, Joo Hyeong Jun, SeungMyeong Jeong, Andreas Kraft, Bob Flynn, Miguel Angel Reina Ortega, Laurent Velez, Samir Medjiah, Xavier Piednoir, Wonbae Son, 안일엽

Our system is to provide services by establishing an autonomous cooperative system of unmanned vehicles.

1,045 Views 3 Respects



[oneM2M Smart Crossroad with NodeMCU \(ESP8266\)](#)

Elisa Nardi, Mattéo GERLIN, Chitong UNG, Léo Delmas, Alexis GIBERT, Samir Medjiah, Andreas Kraft, SeungMyeong Jeong, Bob Flynn, Miguel Angel Reina Ortega, Laurent Velez, Xavier Piednoir, Wonbae Son, 안일엽

The tutorial shows how to integrate a nodeMCU IoT device (having different sensors and actuators) into an **oneM2M** IoT Platform.

821 Views 3 Respects



[oneM2M Hackathon Team Green Cane 2024](#)

Kyeonghwan Nam, 황덕연, 이은진, 권수빈, 홍찬희, Bob Flynn, Andreas Kraft, SeungMyeong Jeong,

Miguel Angel Reina Ortega, Poornima Shandilya, JaeSeung Song

For the safety of the visually impaired, we create smart white canes with various functions. "Smart White Cane Notification System"

780 Views 12 Respects



[oneM2M based Smart Campus IoT system](#)

Ayu Parmar, Shubham Mante, Ihita G, Andreas Kraft, SeungMyeong Jeong, Bob Flynn, Miguel Angel Reina Ortega, Laurent Velez, Samir Medjiah, Xavier Piednoir, Wonbae Son, 안일엽, aman singh

oneM2M based smart campus IoT solutions including air quality alert system & automated room ventilation for COVID risk minimization.

693 Views 2 Respects



[oneM2M RemoteController/ServoMotor with NodeMCU \(ESP8266\)](#)

Samir Medjiah, Miguel Angel Reina Ortega, Laurent Velez

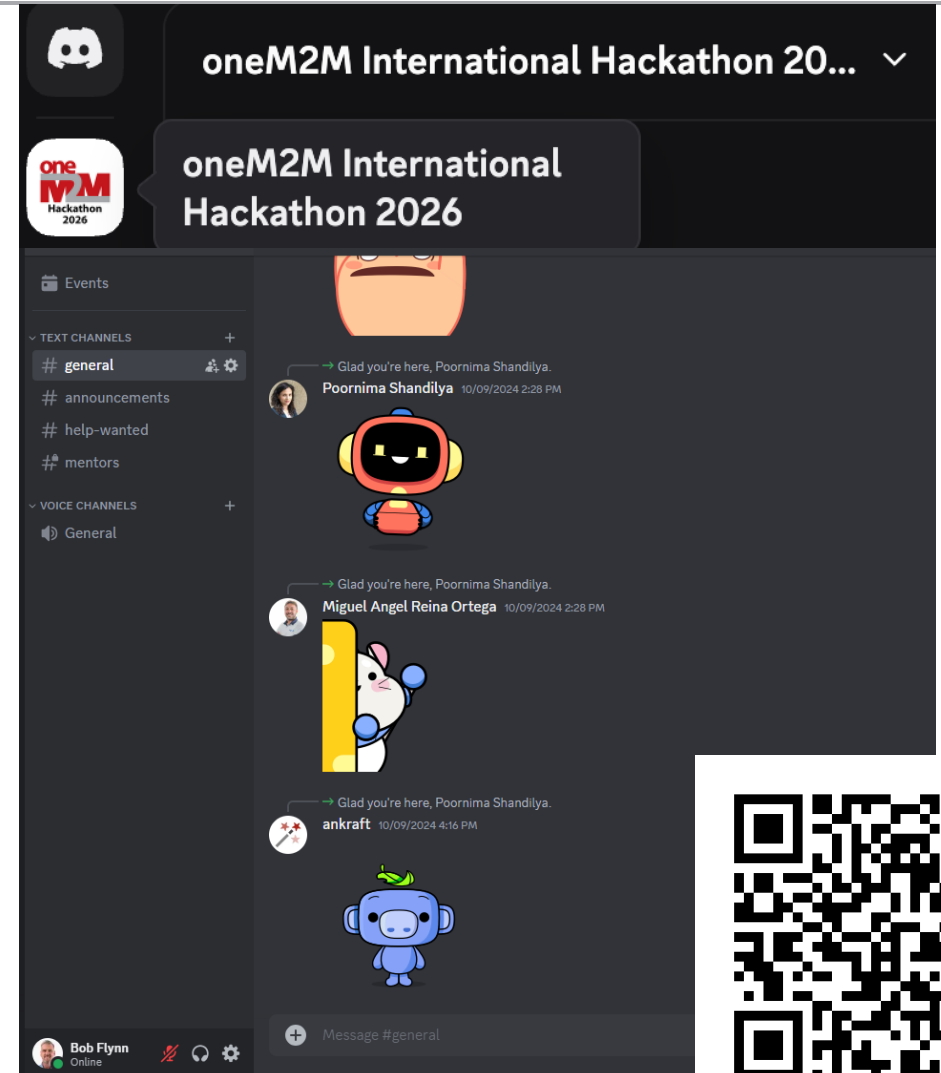
The tutorial shows how to integrate a nodeMCU IoT device (having different sensors and actuators) into an **oneM2M** IoT Platform.

521 Views 0 Respects

Communication tool



- You'll get invited soon
- You should join discord to make this online hackathon **ACTIVE**
- Discord for communication
 - Announcements/Logistics
 - Networking
 - Public/private QnA (mentoring)



Hackathon Mentors

Andreas Kraft (oneM2M & MEC) → ACME

- Over 20 years at Deutsche Telekom in IoT, execution environments, and remote management
- Freelance consultant for IoT and oneM2M topics

Bob Flynn (oneM2M & MEC)

- Chair of oneM2M Test and Developer Ecosystem, oneM2M
- Freelance consultant: oneM2M, R&D for IoT technologies

Miguel Angel Reina Ortega (oneM2M & MEC)

- 17 years at ETSI as testing expert

JaeSeung Song (oneM2M+MEC) → tinyIoT

- TP Vice-chair with many IoT R&D project experiences

Kaleb Gebremeskel (oneM2M+MEC)

- Senior Software Engineer at JLAB (CNIT & AdSP)
- ESTIMED WP6 Manage, oneM2M member, and main contributor to Developer Guide on Interworking between oneM2M and MEC

Poornima Shandilya (oneM2M)

- Lead for the development and deployment of a oneM2M IoT platform, enabling secure and scalable IoT solutions

SeungMyeong Jeong (oneM2M) → Mobius4

- SDS WG Vice-chair with many IoT R&D project experiences
- oneM2M developer tutorials since 2020

Anupama Vattam (oneM2M)

- Chief Technology Architect at the Smart City Living Lab
- Team member for development and deployment of oneM2M IOT platform.

Hemant Magadum (oneM2M)

- Scientist E/ Team lead at CDAC

Andreas Neubacher (oneM2M)

- oneM2M 3GPP standards expert

Massimo Vanetti (oneM2M)

- oneM2M RDM WG Chair

Mudassar Khan (MEC)

- Research Engineer at xFlow Research | MEC, NFV

Muhammad Umair Kahn (MEC)

- Research Engineer at xFlow Research | MEC, NFV & Cloud-Native Systems



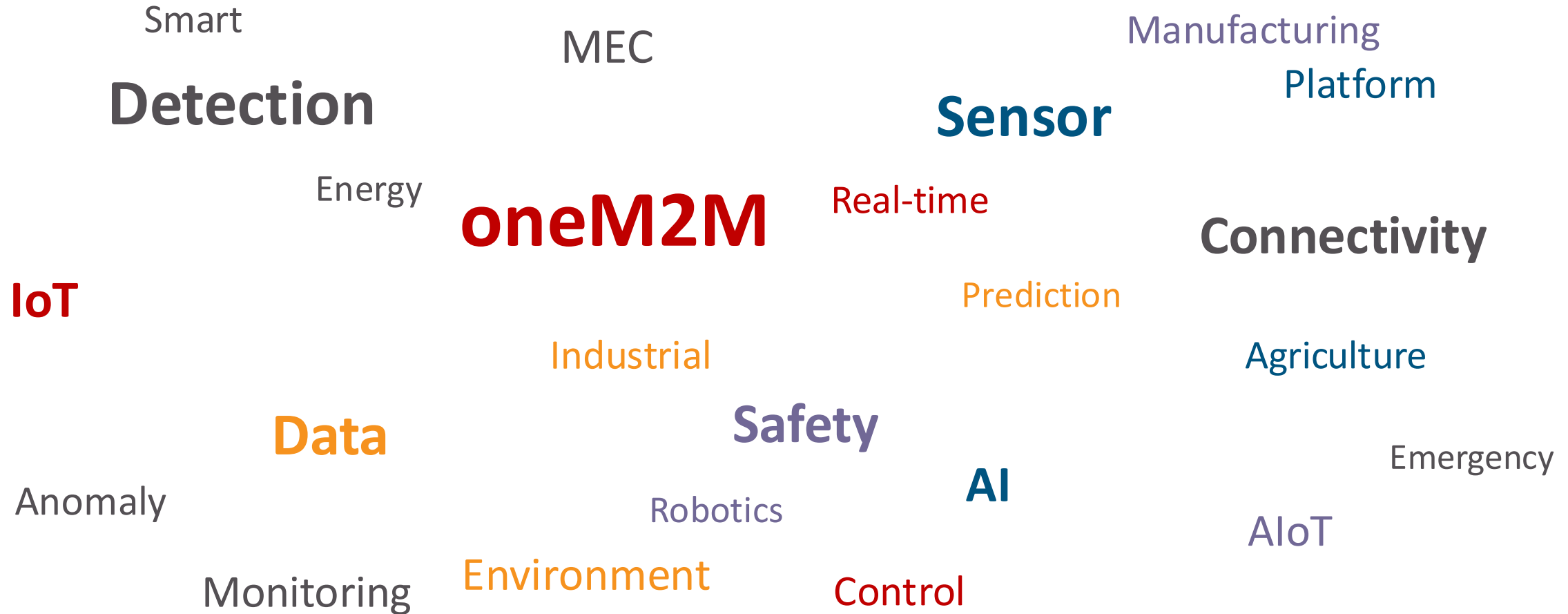
Global Participation Map



32 teams · 9 countries

- South Korea — 16
- India — 8
- United States — 1
- Germany — 1
- Italy — 2
- Austria — 1
- Egypt — 1
- Malaysia — 1
- Nigeria — 1

Project Theme Word Cloud



Participant Teams & Projects 1/6



Country	Team	Project
Austria	FHTW-AIOT	Vehicle Cabin Monitoring and HVAC Control System Using oneM2M Middleware in SDV Architecture
Egypt	Nahed Innovation	HEXA-TPU-RT BDO-SKIN
India	Genesis	oneM2M TrafficResilience - A Standards-Native AIoT Platform for Adaptive Mixed-Traffic Intersections
India	HFCL SiteSense	SafeZone Edge FWA CPE anchored AIoT for Industrial Site Safety and Machine Health
India	INVAS-AG	Smart Farming
India	IT leaders	AI Travel agent

Participant Teams & Projects 2/6



Country	Team	Project
India	RoboMaitri	RoboMaitri
India	Team Nexus	SwarmLink-M2M Autonomous Multi-Robot Coordination and AI Handover over oneM2M and ETSI MEC
India	Teknovate	AIoT for Smart utilities
India	ROSTAN ZeroTime	Predictive maintenance that turns a bearing's vibration into a repair ticket before the machine stops, on oneM2M and ETSI MEC
Italy	JUNIOR TAKA BOUGUIA	JUNIOR TAKA BOUGUIA
Italy	UNIMORE-IngInf-Mantova	IOT INDUSTRIAL - ANAMOLY DETECTION

Participant Teams & Projects 3/6



Country	Team	Project
Malaysia	SignalnSilicon	RescueTwin
Nigeria	Nexalware	GridSense Orchestrator Offline-First, Power-Aware Device Coordination on oneM2M
South Korea	4IoT	oneM2MBased Heavy Equipment Proximity Hazard Detection System
South Korea	ALOM	Xray product defect detection
South Korea	BounceAgain	CindySense oneM2M-Based Smart Workout Tracking System

Participant Teams & Projects 4/6



Country	Team	Project
South Korea	CCGN	Edge AIbased IoT Device Anomaly Detection System on oneM2M and MEC
South Korea	Crtl sea	Adaptive AIoT Data Continuity Management Service for Maritime Communications Based on MEC and oneM2M
South Korea	EdgeGuard	EdgeGuard MECenabled AIoT Collision Prediction Service for WorkerForklift Safety
South Korea	EdgeWay	Floor Anomaly Detection and Robot Operation Management Service Based on MultiRobot Driving Data
South Korea	Even Bubble lock is rock	Privacy Bubble Keyring

Participant Teams & Projects 5/6



Country	Team	Project
South Korea	FTMLCS.US	LeafSentry
South Korea	IEUM	AIoT Based Flood Risk Management and Edge Based Safety Control System for Underground Infrastructure
South Korea	Jaeho Lee	A low cost, low power road surface icing monitoring service based on oneM2M
South Korea	Just do it	PinchGuard AIoTBased Industrial PinchPoint Accident Prevention System
South Korea	Machine Pulse Lab	QuietFix

Participant Teams & Projects 6/6



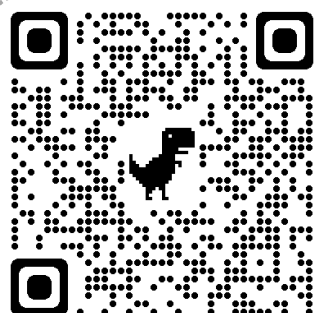
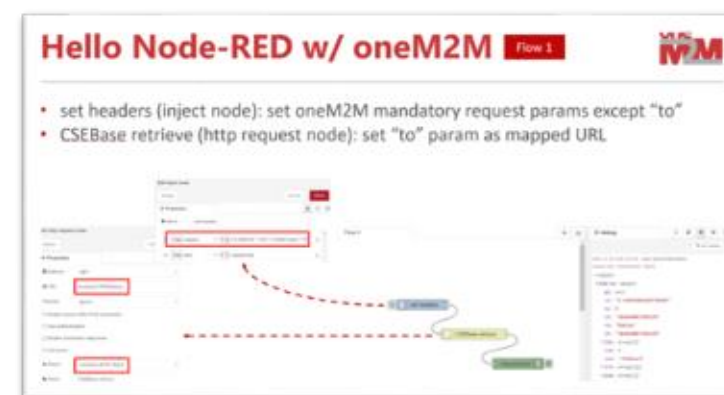
Country	Team	Project
South Korea	openSoom	openSoom AIoT Service for DataDriven Indoor Air Purification and Ventilation Management
South Korea	PolaGrid	PolaGrid
South Korea	twom2m	Intelligent Fire Evacuation Service
United States	Nittany Nerds	gRPB-M2M
Germany	Aadhya	oneM2M device context that follows the device across federated ETSI MEC edges

Teaching materials

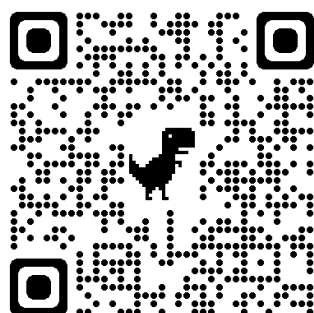


- oneM2M Recipes

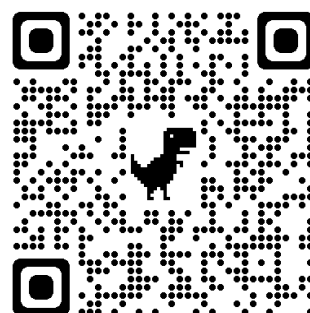
<https://recipes.onem2m.org/>



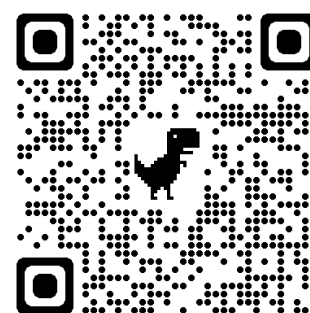
oneM2M
Developers
Tutorial 2020
(ENG, KETI)



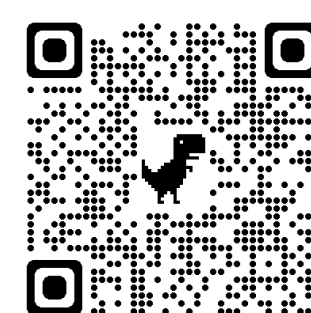
oneM2M
Developers
Tutorial 2021
(KOR, KETI)



oneM2M Developers Tutorial 2022
(basic and advanced courses)
(KOR, KETI)



MOOC
(ENG, C-DOT)



Tutorial w/
Jupyter
Notebook
(ENG, DT)

Evaluation criteria – The Rubric



- Innovative Use of oneM2M
- Security and Privacy
- Data Interoperability and Discoverability
- CSE Handover & Service Continuity (TR-0077)
- oneM2M–MEC Application (TR-0077)
- VM Deployment (TR-0086)
- Federated Learning (TR-0084,TR-0077)
- Geo-location Services
- MEC Services (TR-0077)
- AI Agents & MCP Integration

Criteria	Proficiency Level		
	Expert (4-5)	Proficient (2-3)	Emerging (0-1)
Innovative use of oneM2M	used 1 or more complex services such as group, announcement, pollingChannel, ...	use 1 or more complex features/attributes of basic resources: TimeSeries, FlexContainer, expiration times, max number of instances, etc.	used basic AE registration, containers, subscriptions
Use of appropriate communication protocols	Devices and applications use multiple different protocols, depending on the communication conditions. For example devices might use different protocols than applications.	Devices and applications use other appropriate protocols for communication. Team describes why the different protocol is appropriate.	All devices and applications use http as the protocol for communication to the oneM2M CSE.
Innovative scenarios	complex device to CSE to multiple application (complex application) IoT: Use oneM2M service functions instead of implementing functionalities on device or in applications	multiple simple devices (e.g. sensor + actuator) to CSE to application IoT: Use oneM2M as a hub to manage communication between applications and devices.	single simple device (e.g. temp sensor) to CSE to application. IoT: Use oneM2M as a "database" to store sensor data
Quality of Article	Includes all basic requirements (in box to the right) plus additional insights. Highlights benefits of using oneM2M.	Includes all basic requirements: system/service overview, H/W, oneM2M resource design (if possible), oneM2M protocol usage (if possible) having pics, diagrams, texts	Included all of the basic requirements with basic formatting and basic graphics. Limited discussion about oneM2M.
Quality of Video	Above average video production (not looking for expert quality, just a reasonable effort) that is enjoyable to watch and makes you want to continue reading the article (if it is at the beginning) or did a good job of tying things together (if it is at the end).	Used props (real, simulated, or virtual) to demonstrate the scenario of the project. Some basic video effects. Generally something you would normally watch to completion.	Basic video showing breadboard or experimental components. Decent to poor quality audio or lighting. Generally something you would not continue watching if you came across this on YouTube.
Quality of Github	Stand alone repository. Gives enough details to build, deploy and navigate source code directories. Can get a good understanding of the purpose of the repo without needing to read the hackster.io article, but includes a link to the article as well.	Basic README included. Build instructions included.	Bare minimum github repository with all source code (or some missing source code, based on visible inspection. i.e. it is clear that some files are missing)
Promotion of oneM2M	Article/video/repository represents a well written advertisement and endorsement of oneM2M.	Refers to oneM2M throughout the article, includes links to oneM2M resources and implementations.	Mentions oneM2M in title and introduction, but does not really direct a reader to oneM2M
Judge points	For interactions or discussions held in public forums (slack , github, etc, but not private emails) that demonstrated some aspects that you found beneficial or demonstrated good grasp of oneM2M.	For interactions that may have occurred in private channels (emails, phone calls) where team solicited support that demonstrated an attempt to gain greater understanding of oneM2M. e.g. can you explain the concepts of pollingChannel, group or announcement.	No interactions during the hackathon beyond administrative questions or configuration issues.
Interview points	In addition to the criteria to the right, team members can describe the methodology used to integrate oneM2M. The team members can describe why they would choose to use oneM2M to continue their project.	In addition to the criteria to the right, the team member can describe the benefits of using oneM2M.	Team members describe the resource tree structure. Team cannot explain the benefit of using oneM2M in their project.



rubric link

oneM2M International Hackathon (2021)





제5회
**KETI 모비우스
국제 개발자 대회**

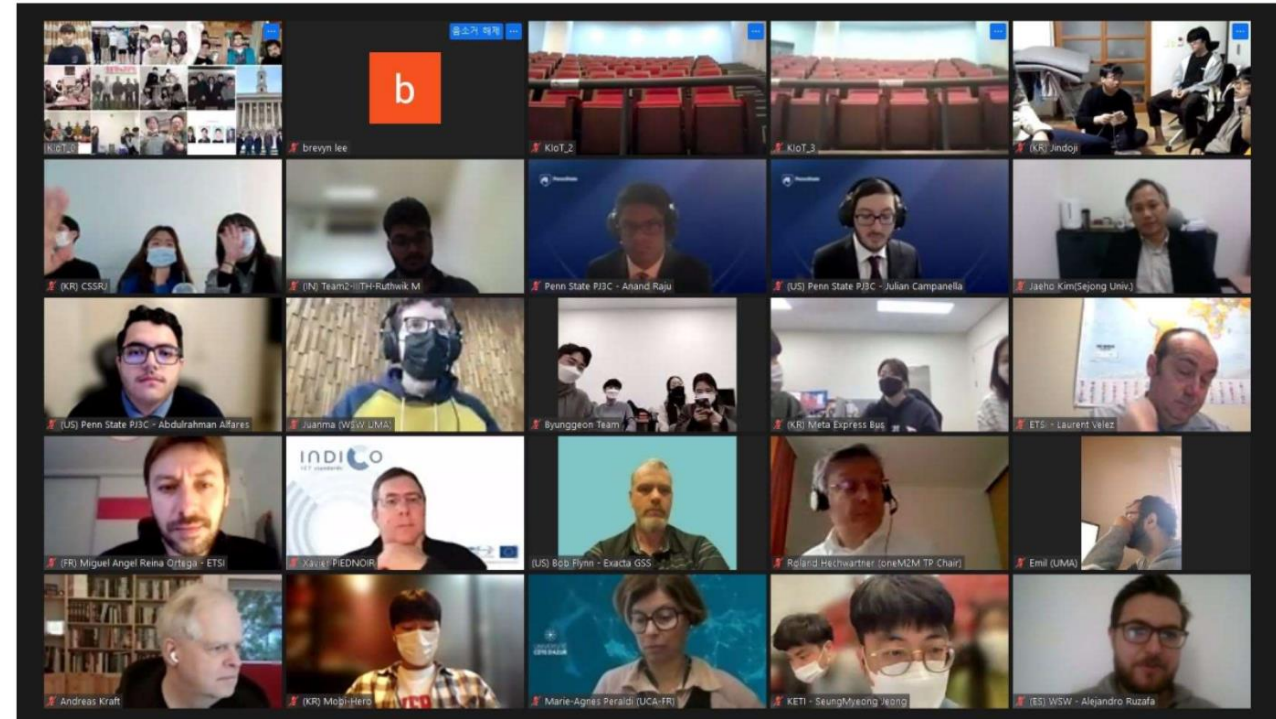
□ 신청 자격 : 제한 없음
□ 신청 기간 : 2021. 9. 10(금) ~ 2021. 9. 23(목)

주 관  한국전자기술연구원
Korea Electronics Technology Institute

후 원  한국지능형사회안전기술협회
 한국정보통신기술협회
 한국정보통신기술협회

대회 개요

- 목적
 - 모비우스 3.0 플랫폼을 활용한 IoT 응용서비스 아이디어 발굴
 - ※ 모비우스 : IoT 표준(oneM2M)을 기반으로 KETI가 개발한 오픈소스 플랫폼
 - 실제 아이디어 구현을 통한 글로벌 사용자 저변확대 및 모비우스 인프라 활용촉진
- 신청대상 : 제한 없음
 - ※ 국내 참가팀은 10-15개팀 내외 선정 예정
 - ※ 해외 참가팀은 미국, 프랑스, 인도 등 모집 중 (10-20개팀 내외)
- 대회 공통 언어 : 영어 (리오프 및 중간점검 미팅, 최종평가, 시상식)
 - ※ 국내 참가팀은 '아이디어 공모 심사, 다제까지마 하군어루 치행'



<=

Video of the Hackathon
Awards Ceremony

Winners

1st Prize: oneM2M Autonomous cooperative smart delivery system (KR)

2nd Prizes: Cellular IoT Irrigation System (US)
A smart automatic pet feeder (CN)

2025 Hackathon Projects

- AI is the main topic for the 2025 hackathon



Hardware Setup
Circuit Diagram and Components

Hardware Setup Block Diagram

- Raspberry Pi 5 — central controller (runs sensor polling, HTTP server/clients, local logic, and cloud communication).
- Prona PM sensor (particulate matter) — serial / TTL interface (measures PM2.5 / PM10).
- Prona CO₂ sensor — CO₂ ppm sensor (likely UART/serial).
- DHT22 — digital temperature & humidity sensor (single-wire digital/data pin).
- 4-Channel Relay Module — switches mains AC loads (fan, lights, AC compressor, and camera power).
- AC Loads — Fan, Lights, AC, Camera, each powered from building AC, switched via relay contacts.
- Power — 5 V (or Pi-compatible) supply for Pi and sensors; separate mains supply for AC loads. Common ground for the DC side.
- Network — Wi-Fi module (or Pi's onboard Wi-Fi) provides connectivity to oneM2M (HTTP).

Team (Country)	Project Name	Highlights
Twinkle (KR)	Twin Link CampusEMS	AI, Digital Twin, Isaac Sim
My Little Soda Farm (KR)	AIoT Mini Farm Platform using oneM2M	AI Inference, Unity
Pringles (KR)	Watch out	AI analytics
GrowMate (KR)	Smart Pot for Single Person Households	Vision AI
9P (KR)	All In One Smart Aquarium	Vision AI
Myong Jung Hyun (KR)	Rehome Lab	IoT automation
Nexus (IN)	Smart Edge AI for Energy-Efficient and Resilient Building Management	AI agents
Eagle Eye (IN)	Smart Crowd Monitoring and Evacuation Using IoT and oneM2M with 5G Connectivity	5G, AI-based risk prediction
Team Echo (IN)	Integrated multi domain monitoring net zero objective	Wi-SUN, AI-driven insights
Penn State Nittany Lions (US)	Cellular based IoT using oneM2M	low-power IoT handover
Tuber Divers (US)	AI For Tubing Outer Diameter Wavy Condition Detection	AI model
VibeTribe (AT)	Workspace Mood Monitor	Real-time analytics
Lorry System (ML)	AI-powered Transport Management System	AI optimization, 5G

oneM2M

Hardware Setup Block Diagram

- CO₂, PM, and DHT22 sensors send real-time environmental data to the Raspberry Pi 5 inside an IP65 box.
- The Raspberry Pi 5 processes the data and sends it to the oneM2M cloud using HTTP POST.
- The cloud can issue control commands back through subscriptions.
- The Raspberry Pi receives these commands and operates a 4-channel relay module.
- The relay controls key appliances such as the fan, lights, AC, and Camera.
- The system provides environment monitoring + remote automation using oneM2M.

Microcontroller System Architecture for IoT Nodes

This diagram illustrates how a microcontroller manages sensor data and controls devices.

- The **Input block** reads sensor values through analog/digital interfaces.
- The **Communication module** connects to the network using the **Wi-Fi module**.
- An internal **HTTP Server** handles incoming requests and parses commands.
- The **Security module** manages hashing, encryption, and secure data handling.
- Based on the received data or commands, the **Action Control** unit determines the next course of action.
- The **output block** sends control signals to the **Relay Control**, which operates external devices.



Any Questions?